

MADHESH P D

Full Stack Developer | AWS Cloud & Backend Engineer

Trichy, Tamil Nadu | +91 8668002081 | madhesh.suguna76@gmail.com | linkedin.com/in/madhesh | github.com/madheshduraismy
| [Portfolio](#)

PROFESSIONAL SUMMARY

Full Stack Developer with hands-on production experience building and deploying scalable, cloud-native web applications on AWS. Specialises in Node.js and TypeScript backend services, React.js single-page applications (SPAs), and serverless architectures using AWS Lambda, API Gateway, Amazon Cognito, SQS, and DynamoDB. Comfortable owning the complete delivery lifecycle — from database schema design and RESTful API development to CI/CD deployment and CDN optimisation — across four concurrent SaaS products in a fast-paced startup environment. Experienced with Agile delivery, role-based access control (RBAC), JWT authentication, microservices, and cross-functional code reviews.

PROFESSIONAL EXPERIENCE

Full Stack Developer | **Packforce Pvt Ltd** | Tiruchirappalli

March 2025 - Present

- Engineered Node.js and TypeScript RESTful backend services for 4 live SaaS production applications (Business Management System, WhatsApp Messenger, Gold Pledge & Repledge, Restaurant Table Booking) on AWS EC2, implementing environment-based configuration and auto-scaling to maintain 99%+ uptime across development, staging, and production environments.
- Architected serverless REST APIs with AWS Lambda and API Gateway, eliminating fixed server costs and enabling on-demand scaling for business-critical workflows; reduced infrastructure overhead by replacing always-on EC2 instances for suitable endpoints.
- Deployed React.js single-page applications (SPAs) across all 4 SaaS products via AWS Amplify CI/CD pipeline, S3 static hosting, CloudFront CDN, and Route 53, achieving faster global page load times through edge caching and automated deployments on every merge.
- Designed multi-database architectures combining PostgreSQL (billing, loans, and orders) with Amazon DynamoDB (high-throughput, low-latency operations), selecting the right storage engine per workload rather than defaulting to a single solution.
- Implemented Amazon Cognito for user authentication, JWT-based authorisation, and role-based access control (RBAC) with OAuth 2.0 flows across all 4 platforms, securing multi-tenant data isolation between user roles.
- Integrated Amazon SQS message queuing to decouple bulk WhatsApp notification delivery, enabling asynchronous, fault-tolerant processing and preventing bottlenecks under high-load scenarios.
- Collaborated in Agile sprint cycles — participating in planning, code reviews, and retrospectives — ensuring consistent delivery cadence across concurrent products.

Web Developer Intern | **Code Bind** | Tiruchirappalli

June 2023 - July 2023

- Built 3 full-stack applications (Car Rental System, Task Management, Tour Booking) end-to-end: UI design, RESTful API development, and database integration using React.js, Node.js, Express.js, and MySQL/PostgreSQL.
- Designed JWT-based authentication and admin dashboard features covering real-time vehicle availability, booking management, task assignment, and itinerary browsing.
- Participated in Agile code reviews and maintained Git version control practices across all repositories, adhering to branching and pull-request workflows.

PROJECTS

Task Management System

Aug 2024 - Oct 2024

- Full-stack application with React.js frontend (Netlify) and Node.js/Express.js backend (Render), enabling real-time task tracking across distributed teams with sub-200ms average API response times.
- Built a dual-role permission system — managers create, assign, and monitor tasks via a dashboard; team members view, update, and complete their assigned work — enforcing data isolation through RBAC.
- Developed CRUD-compliant RESTful APIs with Express.js and a relational PostgreSQL database, covering user profiles, task records, assignments, priorities, deadlines, and status workflows.

- Implemented JWT authentication with session management supporting both roles, ensuring secure, stateless authorisation across all API endpoints.

Agri-Bot - AI Agricultural Assistant

Nov 2024 - Feb 2025

- Built an AI-powered assistant integrating with NPK sensor hardware to capture real-time soil pH, nitrogen, phosphorus, and potassium readings for data-driven crop recommendations.
- Developed a machine learning (ML) crop recommendation engine using Python, TensorFlow, and scikit-learn, analysing sensor readings to suggest optimal crops for detected soil profiles and reducing crop failure risk for small-scale farmers.
- Delivered a unified Python full-stack platform connecting the NPK data pipeline, ML recommendation logic, and a government agricultural scheme discovery module into a single interface.

Car Rental System

Jan 2023 – Mar 2023

- Full-stack car rental platform featuring real-time vehicle availability, secure online payments, and a comprehensive admin dashboard, resulting in a 20% increase in online bookings.
- Engineered backend APIs for booking management with Node.js and Express.js, integrated a cloud-hosted PostgreSQL database, and implemented JWT authentication for secure user sessions.

TECHNICAL SKILLS

Frontend: React.js, JavaScript (ES6+), TypeScript, HTML5, CSS3

Backend: Node.js, Express.js, RESTful API Design, Serverless Architecture, Microservices, NestJS (familiar)

AWS: EC2, Lambda, API Gateway, Amplify, S3, CloudFront, Route 53, Cognito, SQS, DynamoDB, IAM, RDS

Databases: PostgreSQL, MySQL, MongoDB, Amazon DynamoDB

Security: JWT Authentication, OAuth 2.0, Role-Based Access Control (RBAC)

DevOps & Tools: Git, GitHub, CI/CD (AWS Amplify), Postman, Linux

Testing: Postman, Manual API Testing

Methodologies: Agile, Sprint-based Delivery, Code Review, REST API Design Principles

EDUCATION

B.E. - Computer Science and Engineering (First Class)

2021 - 2025

Hindusthan Institute of Technology, Coimbatore | CGPA: 7.21

HSC — 87%, Lakshmi Matric Higher Secondary School

2021

SSLC — 83%, Lakshmi Matric Higher Secondary School

2019